

Lesion-Wise False-Positive Reduction for Brain Tumor Candidate Masks in Multi-Modal MRI via Mask-Gated 2.5D Spatiotemporal Classification

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Abstract—Automated detection systems for intracranial tumors face a critical trade-off between high sensitivity and a manageable false-positive (FP) burden. High-sensitivity segmentation models can detect small lesions but often produce numerous spurious candidates from vascular mimics or imaging artifacts, contributing to clinical “alarm fatigue.” We present a clinically aligned coarse-to-fine cascade: a high-sensitivity Stage-0 segmentation candidate generator (implemented with nnU-Net in this study) followed by a dedicated 2.5D spatiotemporal classifier operating on multi-modal MRI patches from T1-weighted contrast-enhanced (T1W+C) and T2-weighted (T2W) scans to filter candidates. A central component is Mask-Gated Modulation, which uses the candidate ROI mask as an explicit spatial attention gate to multiplicatively modulate intermediate features, amplifying lesion-specific texture while suppressing background confounders rather than relying on passive mask concatenation. We evaluate the framework on a retrospective cohort from two medical centers (>12,000 subjects), with a prevalence-aware held-out test set and lesion-wise metrics. Controlled ablations show consistent gains over image-only and early-fusion baselines. At the primary operating point ($\tau = 0.05$), we achieve 91.93% lesion-wise sensitivity with 0.82 FP/scan, corresponding to a 46.4% FP reduction relative to the Stage-0 generator. An adjustable decision threshold further enables explicit sensitivity–FP trade-offs for screening versus high-precision use.

Keywords—Brain tumor detection, false positive reduction, mask-gated attention, cascaded deep learning, MRI analysis.

I. INTRODUCTION

Automated detection of intracranial tumors (glioma, meningioma, and brain metastases) is pivotal for treatment planning [1]. Although deep learning segmentation models such as nnU-Net [2] have achieved high sensitivity, their clinical utility is often compromised by excessive false positives (FPs) [3], [4]. In routine practice, persistent “false alarms” (e.g., vascular mimics) can induce alarm fatigue and reduce diagnostic efficiency.

To address this, we implement a coarse-to-fine cascaded framework: a Stage-0 generator prioritizes sensitivity, followed by a Stage-1 refinement classifier. Unlike prior works that focus on balanced datasets, this study evaluates the framework on a prevalence-aware test set curated from two medical centers, reflecting the true clinical distribution of tumor types.

The central question is how to effectively leverage candidate masks for classification. We hypothesize that Mask-Gated modulation, previously explored in prior medical imaging models [5], is superior to simple concatenation for this task. By spatially gating features, the network focuses on lesion textures while suppressing background noise.

Contributions: This paper presents a comprehensive evaluation of a cascaded lesion detection framework. Our primary contributions are:

- **Application of Mask-Gated FP Reduction:** We adapt multiplicative mask-gated feature modulation to filter brain tumor candidates, suppressing anatomical mimics beyond passive mask concatenation.
- **Clinically Realistic Evaluation:** We validate on a two-center retrospective cohort using a prevalence-aware test protocol to reflect non-uniform clinical tumor distributions.
- **Tunable Clinical Operating Points:** An adjustable probability threshold enables explicit sensitivity–FP trade-offs across screening and high-precision use cases.
- **Lesion-wise Optimization:** Training and evaluation are lesion-wise, leveraging hard negatives from healthy scans to improve robustness against confounders.

II. RELATED WORK

A. High-Sensitivity Candidate Generation

The U-Net architecture [6] and its self-configuring variant, nnU-Net [2], have become the de facto standards for biomedical image segmentation. While highly effective at delineating boundaries, these models are typically optimized using voxel-level loss functions (e.g., Dice loss), which do not explicitly penalize lesion-level FPs. In small-object detection tasks, such as metastasis detection, this often results in numerous small, spurious predictions [7]. Consequently, in our framework, nnU-Net is utilized solely as a high-sensitivity candidate generator, shifting the burden of specificity to a dedicated downstream classifier.

B. False Positive Reduction in Cascaded Frameworks

Cascaded or “coarse-to-fine” architectures are widely adopted in medical imaging when high sensitivity is non-negotiable. Standard approaches typically involve extracting 2D or 3D patches around candidate centers and classifying them using a secondary CNN. Techniques such as hard negative mining, where the second stage is trained on difficult FPs generated by the first stage, have proven effective in lung nodule detection [8]. Our work extends this paradigm to multi-class brain tumor detection, emphasizing the processing of 2.5D (multi-slice) spatiotemporal features to capture volumetric context efficiently.

C. Segmentation-Guided Attention Mechanisms

Segmentation-derived priors have been increasingly used to guide downstream classification. Beyond simple mask concatenation, attention-gated designs learn spatial modulation to suppress irrelevant context and highlight salient regions [5]. Such mechanisms appear in medical segmentation (e.g., Attention U-Net [9]) and generic recognition models (e.g., Residual Attention Networks [10]). Relatedly, segmentation pathways can provide regional priors for grading tasks [11]. Building on these ideas, our Mask-Gated module uses the ROI

mask as an explicit attentional gate to multiplicatively modulate features, focusing on lesion texture while suppressing vascular mimics.

III. METHODOLOGY

A. Overview of the Two-Stage Pipeline

As illustrated in Fig. 1, we follow a clinically aligned coarse-to-fine workflow. In this study, Stage-0 is instantiated using nnU-Net as a high-sensitivity segmentation candidate generator, producing a set of candidate lesion masks from multi-modal MRI while intentionally tolerating a high FP burden to minimize missed lesions. Stage-1 then performs lesion-wise FP reduction: each candidate is converted into a 2.5D patch and classified as true positive (TP) or FP, and candidates predicted as FP are filtered out to produce a configurable candidate set for downstream clinical reading (Fig. 1). Importantly, Stage-1 uses an adjustable decision threshold τ to support clinically configurable operating points that balance lesion-wise sensitivity against FP burden.



Fig. 1. Clinical coarse-to-fine workflow for intracranial tumor screening.

B. Candidate Labeling for False Positive Reduction

To train the FP reduction classifier, each Stage-0 candidate c_i is assigned a ground-truth label $y_i \in \{0, 1\}$ based on its spatial overlap with the expert-annotated ground truth mask M_{GT} . We define the volumetric Intersection-over-Union (IoU) as:

$$IoU(c_i, M_{GT}) = \frac{|M(c_i) \cap M_{GT}|}{|M(c_i) \cup M_{GT}|} \quad (1)$$

where $M(c_i)$ denotes the binary mask of the i -th candidate. A candidate is labeled as:

- **True Positive (TP, $y_i = 1$):** if $IoU(c_i, M_{GT}) \geq \delta_{TP}$, where $\delta_{TP} = 0.1$.
- **False Positive (FP, $y_i = 0$):** if $IoU(c_i, M_{GT}) < \delta_{TP}$.

Crucially, for negative MRI studies (scans with no tumors), M_{GT} is an empty set. Consequently, all candidates generated by Stage-0 on negative scans are labeled as FP ($y_i = 0$). This “Hard Negative Mining” strategy [8] is essential to suppress false alarms in healthy subjects, reflecting real-world clinical screening requirements.

C. 2.5D Patch Construction

For each candidate c_i , we extract a 2.5D patch centered on the candidate's geometric centroid. For negative scans, these centroids are strictly derived from the FP candidate masks erroneously generated by Stage-0. This ensures that the Stage-1 classifier is explicitly trained to discriminate the most challenging confounding structures, thereby performing hard negative mining, rather than evaluating randomly sampled healthy brain tissue.

The input tensor X_i is constructed from the two image channels (I_{T1W+C} , I_{T2W}) and, depending on the ablation setting, the binary ROI mask $M(c_i)$. The classifier operates on tensors of shape (C, D, H, W) , where $D = 15$ represents the through-plane depth (slices), and spatial dimensions are resized to $H \times W = 128 \times 128$. This 2.5D approach balances the capture of volumetric context with computational efficiency compared to full 3D processing.

D. Mask-Gated Modulation

We hypothesize that the ROI mask should actively guide feature extraction rather than serving as a passive input. Let $x_{img} \in \mathbb{R}^{2 \times D \times H \times W}$ be the image tensor and $x_{mask} \in \mathbb{R}^{1 \times D \times H \times W}$

be the corresponding binary mask. We introduce a Gating Branch $G(\cdot)$, consisting of a convolutional layer followed by a Sigmoid activation, to transform the binary mask into a soft attention map. The feature modulation is formulated as:

$$X_{modulated} = x_{img} \odot (1 + \alpha \cdot G(x_{mask})) \quad (2)$$

where $\odot$ denotes element-wise multiplication and α is a scaling factor (set to 0.5 in our experiments). This multiplicative mechanism enhances the signal of texture features within the candidate region while suppressing background noise. The modulated tensor is then passed to the backbone network.

E. Backbone Classifier

The classifier utilizes an R(2+1)D-18 backbone [12], which factorizes 3D convolutions into separate 2D spatial and 1D temporal convolutions. This architecture is chosen for its well-suited ability to model anisotropic medical data (where slice thickness differs from in-plane resolution). The network is initialized with weights pretrained on Kinetics-400 [13]. To adapt to our multi-modal input, the first convolutional layer weights are averaged and expanded to match the input channel dimensions. The final fully connected layer outputs two logits corresponding to the TP and FP classes.

F. Thresholded Inference

During inference, the Stage-1 classifier outputs a TP probability $p_i = \text{Softmax}(\text{logits})$. Candidates are retained only if they satisfy the decision rule:

$$\hat{y}_i = \begin{cases} 1(\text{Retain}), & \text{if } p_i \geq \tau, \\ 0(\text{Discard}), & \text{if } p_i < \tau, \end{cases} \quad (3)$$

The threshold τ serves as a configurable operating point. Lowering τ prioritizes sensitivity (screening mode), while raising τ reduces the FP burden (diagnostic mode).

IV. EXPERIMENTS

A. Dataset and Clinical Prevalence

This retrospective study curated paired multi-modal brain MRI scans, including T1-weighted contrast-enhanced (T1W+C) and T2-weighted (T2W) sequences, from two institutions under Institutional Review Board (IRB) approval. The held-out test cohort was constructed via case-level stratified sampling to approximate epidemiological prevalence patterns (Central Brain Tumor Registry of the United States (CBTRUS)) while

maintaining a balanced evaluation across tumor entities [14]. Although brain metastases can be substantially more frequent than primary brain tumors in clinical practice (reported up to $\sim 10\times$) [15], we did not enforce a $10\times$ sampling ratio; instead, metastasis cases were included as the largest single subgroup to reflect clinical screening burden without allowing one diagnosis to dominate test-set performance. To rigorously evaluate specificity against FPs, imaging-confirmed negative scans were also included.

As detailed in Table I, the dataset comprises 12,437 subjects (12,188 training; 249 testing). Since our framework operates at the lesion level, the extracted cohort yields 15,398 annotated lesions for training and 223 evaluable lesions for the independent test set.

TABLE I. DATASET STATISTICS (SUBJECTS AND LESIONS)

Category	Train		Test	
	Subjects	Lesions	Subjects	Lesions
Neuroma	2,191	2,252	18	18
Meningioma	4,001	4,341	36	41
Brain Metastasis	3,347	7,009	42	121
Glioma	778	1,099	15	25
Pituitary Adenoma	761	697	12	18
Negative	1,110	-	126	-
Total	12,188	15,398	249	223

Note— A patient may have multiple lesions.

B. Stage-0 Candidate Generator

Stage-0 uses nnU-Net in full-resolution configuration with patch size $96 \times 96 \times 96$. Stage-0 serves solely as a high-sensitivity candidate generator to provide the initial set of ROIs C . The nnU-Net model was trained using the training cohort only and was used purely for candidate generation.

C. Data Partitioning for Patch Classification

For the Stage-1 classification task, candidates identified by Stage-0 were extracted as 2.5D patches. The independent test set was first separated at the case level, ensuring that no scans used for testing appeared in training or validation. Within the training cohort, we performed a patch-level split to construct the train/validation sets, because a single MRI case may contain both TP candidates (tumors) and FP candidates (mimics), and forcing strict case-level grouping would unnecessarily reduce the diversity of hard negatives. Thus, each candidate patch is assigned to exactly one split, preventing duplicate patches from appearing in both training and validation. The resulting Stage-1 cohort consists of 21,040 patches for training/validation and 590 independent patches for testing (Table II).

The independent test set originally comprised 223 lesions. Stage-0 successfully detected 209 of these lesions, which were subsequently extracted as true positive (TP) patches. The remaining 14 lesions failed to be detected by Stage-0 (constituting false negatives, FN), which directly corresponds to the Stage-0 baseline sensitivity of 93.72% (209/223).

TABLE II. STAGE-1 PATCH-LEVEL COHORT

Split	TP Patches	FP Patches	Total Patches
Train	9,950	11,090	21,040
Val	2,487	2,773	5,260
Test	209	381	590

D. Implementation Details

The framework was implemented in PyTorch. Training was conducted on an NVIDIA Tesla V100 GPU.

TABLE III. TRAINING HYPERPARAMETERS

Component	Setting
Backbone	R(2+1)D-18 (Kinetics-400 pretrained)
Input	2 channels (T1W+C, T2W) + ROI mask for gating
Optimizer	AdamW, LR 1×10^{-3} , weight decay 1×10^{-2}
Scheduler	OneCycleLR (cosine), $pct_start = 0.1$
Epochs / Patience	300 / 30
Batch / Epoch samples	50 / 15,000
Loss	Cross-entropy with label smoothing 0.05
Val TTA	Left-right flip (enabled)

Note—LR: learning rate; TTA: test-time augmentation.

E. Preprocessing

Each 2.5D patch was standardized using a robust per-patch normalization (median/MAD; fallback to mean/std when necessary) with value clipping to mitigate outliers, following common practice [2] for multi-site MRI intensity variability with outlier clipping (1st–99th percentile) for numerical stability.

F. Baselines / Ablation Variants

To isolate the contribution of the Mask-Gated mechanism, we compare three variants under identical training protocols:

- **Mask-Gated:** Uses Eq. (2) to modulate features.
- **ROI-Concat:** The mask is concatenated as a 3rd channel ($x_{in} \in \mathbb{R}^{3 \times D \times H \times W}$). No gating is applied.
- **No-ROI:** The model receives only the 2 image channels. All mask-based information is removed.

G. Evaluation Metrics

We evaluate the framework at two distinct levels: the individual patch classification performance and the end-to-end clinical detection efficacy. To assess the discriminative power of the Stage-1 model, we report lesion-wise classification accuracy on the independent test set. To characterize clinically relevant performance and a proxy for mitigating “alarm fatigue,” our primary system-level metrics include Lesion-wise Sensitivity, Lesion-wise Precision, and False Positives per Scan (FP/scan). FP/scan denotes the average number of false-positive lesions per scan. Furthermore, Free-Response Receiver Operating Characteristic (FROC) analysis is employed to evaluate system robustness and configure optimal clinical operating points by sweeping the probability threshold $\tau \in [0.01, 0.9]$.

V. RESULTS

A. Classification Performance on Lesion Patches

The primary decision threshold $\tau = 0.05$ was determined based on validation set performance, specifically selected as the optimal clinical balance point to ensure a lesion-level sensitivity above 90% while constraining the average FP burden to fewer than one per scan.

The technical discriminative capability of the Stage-1 classifier was first evaluated at the patch level. Table IV summarizes the performance at the primary operating threshold ($\tau = 0.05$). The Mask-Gated model achieved an overall accuracy of 64.75%, correctly classifying 382 out of 590 independent test patches. In terms of TP retention, it preserved the vast majority of true lesions generated by Stage-0, achieving a TP accuracy of 98.09% (205/209 TP patches correctly identified). Meanwhile, its specificity reached a FP accuracy of 46.46%, correctly rejecting 177 out of 381 spurious candidates. By comparison, the baseline variants exhibited less favorable trade-offs: No-ROI achieved 63.90% overall accuracy with 95.22% TP accuracy and 46.72% FP accuracy, whereas ROI-Concat maintained 100.00% TP accuracy but suffered from markedly reduced FP discrimination (16.80%), resulting in substantially lower overall accuracy (46.27%).

TABLE IV. PATCH-WISE CLASSIFICATION ACCURACY

Variant	Overall Acc. (%)	TP Acc. (%)	FP Acc. (%)
No-ROI	63.90	95.22	46.72
ROI-Concat	46.27	100.00	16.80
Mask-Gated	64.75	98.09	46.46

B. End-to-End Clinical False Positive Reduction

In the end-to-end clinical workflow, the primary objective is to reduce the “alarm fatigue” associated with excessive FPs from the high-sensitivity Stage-0 generator. As shown in Table V, the Stage-0 candidates yield 1.53 FP/scan on the test set. With the Mask-Gated Stage-1 filter, FP/scan significantly decreases to 0.82, corresponding to a 46.4% reduction relative to Stage-0. Importantly, this reduction is achieved while maintaining a lesion-wise sensitivity of 91.93%. The ROI-Concat baseline reduces FP/scan to 1.27 but remains less effective than Mask-Gated filtering, suggesting that feature-level gating better suppresses background confounders.

TABLE V. ABLATION STUDY: END-TO-END CLINICAL KPIS

Pipeline Variant	Lesion Sens. (%)	Lesion Prec. (%)	FP/scan	Δ FP	Δ TP
Stage-0 Baseline	93.72	35.42	1.53	—	—
+ Stage-1 (No-ROI)	89.24	49.50	0.82	-178	-10
+ Stage-1 (ROI-Concat)	93.72	39.73	1.27	-64	0
+ Stage-1 (Mask-Gated)	91.93	50.12	0.82	-177	-4

C. FROC Analysis and Model Comparison

The overall system robustness was evaluated via FROC analysis. As illustrated in Fig. 2, the Mask-Gated model consistently dominates the ablation baselines across clinically relevant FP/scan.

A quantitative comparison at fixed clinical budgets highlights this superiority. For instance, at a strict budget of 0.4 FP/scan, the Mask-Gated model achieves 88.61% sensitivity, outperforming the ROI-Concat (85.20%) and No-ROI (78.03%) variants. At our primary operating budget of approximately 0.8 FP/scan, the Mask-Gated model maintains 91.93% sensitivity, while the ROI-Concat and No-ROI models achieve 91.48% and 89.24%, respectively. This gap in the low-FP regime supports the advantage of multiplicative gating over early-fusion concatenation.

Furthermore, across the entire operating range, particularly at the primary clinical operating point ($\tau = 0.05$), the Mask-Gated model achieves a high sensitivity of 91.93% with only 0.82 FP/scan. Notably:

- **Low-FP Regime (< 0.5 FP/scan):** Even under extremely strict FP budgets, the Mask-Gated curve rises significantly faster than the baselines. This indicates that the spatial gating mechanism effectively ranks true positive lesions higher than confounding background structures like blood vessels.
- **Ablation Trends:** The ROI-Concat model (Early Fusion) outperforms the No-ROI baseline, confirming that providing the segmentation mask as a spatial prior is beneficial. However, the consistent gap between the Mask-Gated and ROI-Concat curves proves that *how* the mask is utilized is critical; feature-level multiplicative modulation (Gating) is more effective at resolving difficult FPs than simple channel concatenation.
- **Sensitivity Saturation:** All models converge as the threshold τ approaches 0.01, reflecting the maximum lesion-wise sensitivity imposed by Stage-0 candidate generation. However, the Mask-Gated model reaches this plateau with a much lower FP burden, which is vital for minimizing “alarm fatigue” in a clinical environment.

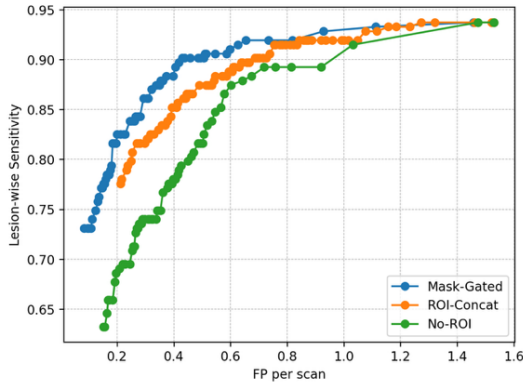


Fig. 2. Comparative Free-Response ROC (FROC) analysis of system-level performance as the decision threshold τ is swept (0.01–0.9); curves correspond to Mask-Gated, ROI-Concat, and No-ROI (see legend).

D. Recommended Operating Points for Clinical Workflows

To further investigate the flexibility of the Mask-Gated framework in various clinical settings, we analyzed the performance of the Mask-Gated model across a range of decision thresholds $\tau \in [0.01, 0.20]$. As summarized in Table VI, the choice of τ allows the system to be tuned to satisfy different clinical priorities:

- **High-Sensitivity Mode ($\tau \leq 0.03$):** At low thresholds, the system prioritizes the retention of every potential lesion. For instance, at $\tau \leq 0.01$, the model achieves its maximum sensitivity of 93.72%. This setting is ideal for initial automated screening where missing a subtle lesion is the primary concern, despite a higher FP burden (1.53 FP/scan).
- **Balanced Clinical Mode ($\tau = 0.05$):** Our selected primary operating point strikes a balance by significantly reducing the FP burden to 0.82 FP/scan while maintaining a high sensitivity of 91.93%.
- **High-Precision Mode ($\tau \geq 0.10$):** As the threshold increases, the system becomes more conservative, effectively filtering out more confounding structures. At $\tau = 0.20$, the average drops to 0.42 FP/scan with a precision increase to 65.79%. This mode is suitable for “confirmatory reading” where the goal is to provide highly confident candidates to minimize physician intervention time.

Overall, increasing τ monotonically reduces FP/scan and increases precision (Table VI), enabling a controllable sensitivity–FP trade-off across screening and confirmatory use.

TABLE VI. QUANTITATIVE IMPACT OF DECISION THRESHOLD (τ) ON CLINICAL PERFORMANCE

Threshold (τ)	Lesion Sens. (%)	Lesion Prec. (%)	FP/scan
0.01	93.72	35.42	1.53
0.03	93.27	42.89	1.11
0.05	91.93	50.12	0.82
0.10	90.58	58.05	0.59
0.20	89.69	65.79	0.42

VI. DISCUSSION

We applied a cascaded framework to address the FP bottleneck in brain tumor detection. Our results confirm that the Mask-Gated mechanism, previously validated in other tasks [5], effectively translates to this domain. By prioritizing textural features within the ROI, it outperforms simple concatenation, particularly in the low-FP regime. The use of a prevalence-aware test set further validates the system’s robustness in realistic clinical scenarios, where class imbalance is inherent.

A. The Efficacy of Mask-Gated Modulation

A central finding of this study is the superiority of the Mask-Gated mechanism over traditional feature fusion methods. While simple ROI concatenation (ROI-Concat) provides a spatial prior, our results show it often leads to conservative

filtering with minimal precision gains. In contrast, the multiplicative modulation in our Mask-Gated model ($1 + \alpha \cdot G(X_{mask})$) acts as a learned spatial attention gate. This mechanism forces the network to prioritize textural features within the candidate volume while actively suppressing confounding background signals, such as vascular enhancement or dural thickening. The FROC analysis (Fig. 2) confirms this dominance, showing that the Mask-Gated approach consistently provides higher sensitivity across all FP budgets, particularly in the critical low-FP regime (< 1.0 FP/scan).

B. Clinical Implications and Alarm Fatigue

In automated screening, “alarm fatigue” caused by excessive FPs is a primary barrier to AI adoption. By reducing the FP burden from 1.53 to 0.82 FP/scan (46.4% reduction), our framework yields fewer than one FP per scan on average. This manageable overhead supports the computer-aided diagnosis (CAD) system functioning as an efficient “second reader,” rather than a source of distraction. Furthermore, our use of hard negative mining, incorporating candidates from tumor-free scans during training, ensures the system remains robust when processing healthy subjects, which is important for large-scale clinical deployment.

C. Implementation Trade-offs and 2.5D Processing

We adopted a 2.5D (15-slice) patch-based approach for the Stage-1 classifier. Compared to full 3D volumes, this configuration significantly reduces the computational footprint while capturing sufficient volumetric context to distinguish true lesions from 2D imaging artifacts. While Stage-1 lost a negligible amount of sensitivity (4 true lesions), the massive gain in precision justifies this trade-off in a high-throughput screening context.

D. Limitations and Future Work

Despite strong performance, certain failure modes persist. Missed lesions were typically very small metastases that failed to trigger the Stage-0 generator. Future iterations could explore more aggressive Stage-0 tuning combined with a 3D Stage-1 classifier to capture even more subtle volumetric cues. Additionally, while this work focused on binary FP reduction, the refined candidate pool generated by Stage-1 provides a high-quality foundation for subsequent differential diagnosis (tumor typing). Expanding the framework to incorporate additional sequences, such as FLAIR or contrast-enhanced T2, could further enhance its ability to characterize atypical pathologies.

VII. CONCLUSION

This paper presented a clinically aligned coarse-to-fine cascade for reducing the FP burden in automated intracranial tumor detection. A high-sensitivity 3D segmentation model is used as a candidate generator, and a subsequent lesion-wise 2.5D spatiotemporal classifier performs FP filtering. By adopting Mask-Gated Modulation, which treats the candidate ROI mask as an explicit attention gate that multiplicatively modulates intermediate features, the proposed filter suppresses vascular mimics and other background confounders beyond passive mask concatenation. On a two-center retrospective cohort ($> 12,000$ subjects), the proposed framework is confirmed

on a prevalence-aware held-out test set and attains a practical primary operating point ($\tau = 0.05$), achieving 91.93% lesion-wise sensitivity with 0.82 FP/scan (46.4% reduction vs. Stage-0). Moreover, the adjustable decision threshold enables explicit sensitivity–FP trade-offs to accommodate different clinical priorities, ranging from high-sensitivity screening to high-precision use. Future work will focus on prospective validation and extending the refined candidate set to downstream tumor typing and longitudinal monitoring.

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